

UGTS-1

Double Verification and Practical Performance Metrics

Restricted geometric-topological query and event substrate

Source-to-spec verification + independent executable verification

34 x 2	93%	12.31 ms	6.07M/s
tests passed two hash seeds	coverage statement + branch	100k pipeline single pinned core	1M workload candidate throughput

VERDICT

GO for a bounded, headless, query/event substrate over controlled relation families. NO GO for universal O(1), zero-memory/zero-power claims, literal physical topology, or hardware throughput claims without a device.

Report issued	15 August 2026
Benchmark runs	14 August 2026 UTC; two complete pinned single-core processes
Source basis	Nine uploaded PDFs; exact hashes in Appendix A
Deliverable	PDF plus reproducible source, tests, metrics, charts, and hardware template

Executive verdict

What survived verification

The corpus becomes technically coherent when read as: typed state -> local spherical support -> compatibility -> relation/event solve -> transition -> lineage/event log. Rendering and hardware remain optional adapters.

The decision in one page

The substrate is internally and computationally valid as a restricted equation-world kernel. The tested reference can evaluate closed state expressions directly, prune by radial-angular support, prevent coupling between incompatible phase/sheet sectors, solve plane and sphere events, route transitions, preserve lineage, and project optional one-bit graphics.

The strongest measured result is not raw graphics speed. It is the semantic separation between frame replay and direct query. Across horizons from 1 to 1000 seconds, scalar direct state evaluation had a fitted time exponent of approximately 0.0009, while 60 Hz replay had an exponent of approximately 0.996. That confirms horizon independence only for the fixed closed trajectory family actually implemented.

On one pinned AMD EPYC virtual core, the complete synthetic pipeline processed 100,000 candidates in 12.31 ms median and 12.80 ms aggregated p99. The 1,000,000-candidate batch took 164.86 ms median at 6.07 million candidates/s. This is practical for bounded headless queries and some 60 Hz workloads near 100k candidates, but it is not yet a million-entity per-frame game loop on one Python CPU core.

What did not survive

- A one-bit flag cannot contain complete continuous world state. It is retained only as route, sheet, parity, validity, or output information.
- A globally signed SDF on a non-orientable surface is not assumed. The implementation uses local quotient charts plus an explicit orientation bit.
- A physical Klein bottle does not form itself because software flips a parity bit. That material claim is rejected.
- Universal $O(1)$, zero memory, zero latency, zero heat, and zero power are rejected. Actual candidate, expression, event-log, and hardware costs remain.
- No liquid-waveguide device was present. Optical loss, joules/event, settling, drift, miss probability, and calibration burden remain unmeasured.

35.4M	76.4M	79.7M	13.7M
support tests/s	compatibility/s	plane roots/s	sphere roots/s
1M batch	1M batch	analytic	analytic

Contents and verification method

Section	Scope
1	Corpus normalization and technical corrections
2	Verified substrate architecture
3	Source-to-spec verification matrix
4	Executable verification and coverage
5	Practical performance metrics
6	Topology, lineage, logs, and projection
7	Hardware measurement boundary
8	Missing work, risks, and kill criteria
Appendix A	Source and build manifests
Appendix B	Reproduction commands and metric definitions

Double verification

1. Source-to-spec: each recurring corpus motif was classified, technically translated, and bounded. Unsupported totalizing claims were not silently imported into the implementation.
2. Spec-to-execution: the translated contract was implemented independently, compiled, tested twice under different Python hash seeds, measured with branch coverage, and benchmarked in two separate pinned single-core processes.

Evidence hierarchy

Evidence class	Meaning
Measured	Two-run CPU timings, memory/RSS, file sizes, residuals, randomized property tests.
Verified in code	State types, support, compatibility, analytic roots, bracketed cross-check, quotient maps, lineage, logs, deterministic output.
Source-derived design	The source corpus motivates the architecture and terminology; it is not independent performance validation.
Conditional	Generic SDF solving, topology beyond quotient routing, large grammar closure, engine integration.
Unmeasured	Optofluidic hardware, energy, loss, calibration, physical reliability.
Rejected	Universal $O(1)$, zero-cost claims, physical self-intersection magic, general AI/biology equivalence.

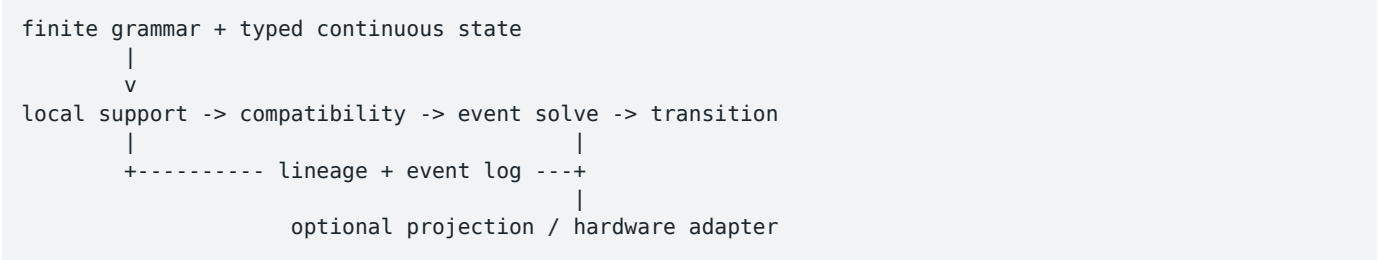
Source discipline

The later corpus documents themselves establish the correct rule: retain implementable operators, demote totalizing claims, and count only verified outputs with error, energy, latency, and calibration boundaries declared.

1. Corpus normalization and technical corrections

Mature reading of the substrate

The decisive corpus shift is from “draw a clever image” to “maintain directly queryable relations, phases, identities, and event surfaces.” Under that reading, a cone or spherical sector is a local support operator; one-bit is routing state; double vacuum is non-coupling between co-located sectors; SDF = 0 is an event or boundary relation; and hourglass/Klein language is a routing or gluing abstraction.



Corrections required for technical validity

Correction	Engineering interpretation
Popcount is not numeric equality	19 decimal is 10011 binary and has popcount 3. Removing zero positions produces a three-symbol feature, not a new numeral equal to 19.
One bit is not complete state	The bit selects a route/sheet/parity/validity. Continuous position, time, phase, uncertainty, thresholds, and lineage remain separate.
Log-polar has a singular center	$\rho = \log(r)$ requires $r > 0$. The projection uses epsilon clamping and treats the center as a special cell.
Non-orientable sign needs chart state	A globally consistent signed distance is unavailable without an orientation cover. The model stores orientation explicitly.
Analytic solving is family-specific	Plane and sphere roots are closed form. Generic relations need bracketing, intervals, or numerical root methods.
Topology is routing, not material magic	Möbius/Klein quotient identifications are implementable. A physical object passing through itself is not inferred.
“Feynman vectors” are renamed	Without an actual quantum path-integral model, the graphics operation is treated as phasor/coverage accumulation and dithering.
Memory follows novelty and representation	Closed state can be recomputed, but exogenous events, branch history, calibration, and schemas still consume storage.

2. Verified substrate architecture

Typed state

The reference state is 104 bytes in a packed NumPy layout. It includes lineage, parent, branch, tags, sheet, orientation, parity, flags, time origin, position, velocity, phase, phase rate, and uncertainty. The packed layout is the minimum authoritative state used by the benchmark; it is not a claim that all future domains fit in 104 bytes.

```
q = (lineage, parent, branch, tags, sheet, orientation, parity,
     t0, position[3], velocity[3], phase, phase_rate, uncertainty)

p(t)      = p0 + v * (t - t0)
phase(t) = wrap(phase0 + phase_rate * (t - t0))
```

Local support

Admission uses a radial interval and a cone angle around a normalized local axis. This is the implementable spherical-support operator: local, bounded, and explicit. It does not remap the complete world into polar coordinates.

```
support(p) = [r_min <= ||p-origin|| <= r_max]
             AND [dot(normalize(p-origin), axis) >= cos(half_angle)]
```

Compatibility and double vacuum

Compatibility combines tags, sheet, orientation, and circular phase tolerance. Two entities can occupy the same coordinate and remain uncoupled because their sectors are incompatible. This captures the useful information-architecture content of “double vacuum” without proposing a second physical vacuum.

Relation, event, and transition

```
t* = first t >= t0 such that
      relation(q(t)) = 0
      AND support(q(t))
      AND compatible(q(t))
      AND t <= horizon

event -> immutable record -> relation-specific transition patch
```

Topology and identity

Möbius and Klein behavior is encoded as quotient gluing on a fundamental rectangle with an orientation bit. Lineage uses deterministic parent/branch/generation-derived IDs. Exogenous changes are written to an irreducible event log.

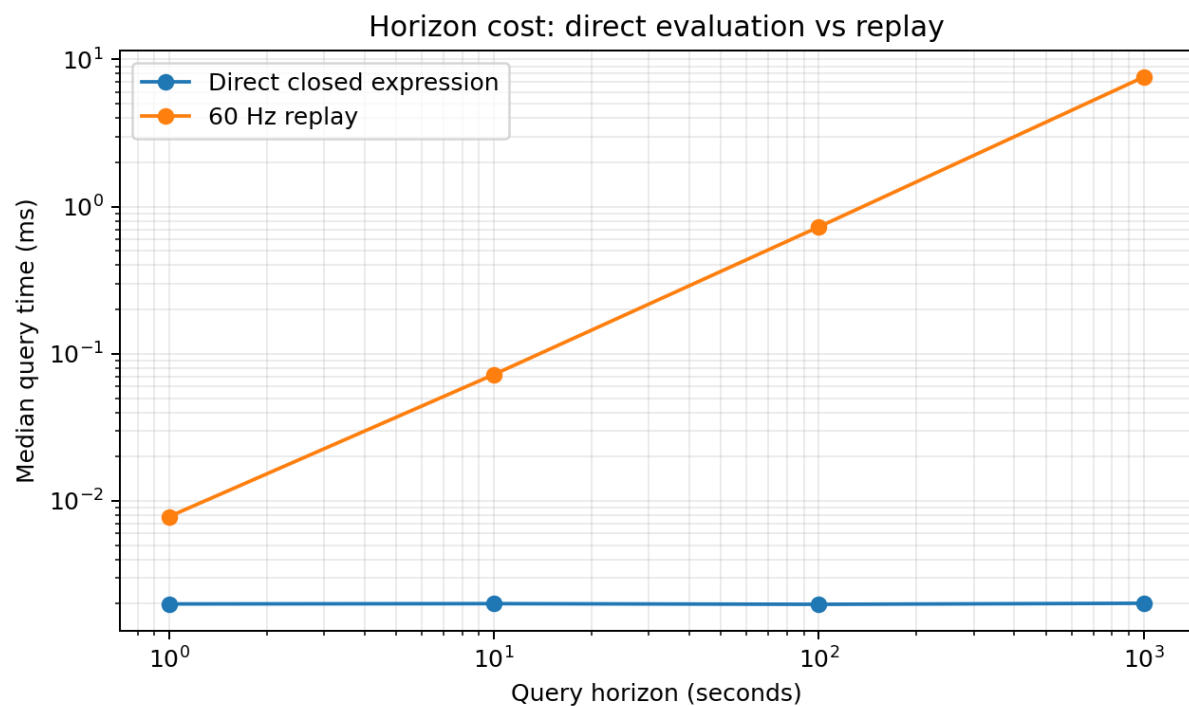


Figure 1. Direct state evaluation remains horizon-independent for the tested closed expression; replay scales with the number of frame steps.

3. Source-to-spec verification matrix

Twenty-five mechanisms were independently classified. “Pass” means the bounded engineering translation is coherent and tested; it does not validate every rhetorical interpretation in the source dialogue.

ID / mechanism	Source pages	Status	Verified result
V01 Radix threshold / digit-count transition	waarom-heeft-19...pdf pp. 1-4	PASS	19 == 0b10011; thresholds occur at b^k .
V02 Active-bit / popcount motif	waarom-heeft-19...pdf pp. 4-7	PASS-WITH-REWRITE	Arithmetic passes; '19 equals 111' is only a named projection.
V03 Pascal parity / Sierpinski structure	waarom-heeft-19...pdf pp. 6-10	PASS	Suitable deterministic fractal grammar; not a proof of unrelated physical claims.
V04 Zero-based offset semantics	21BenBurgers...pdf pp. 1-4	PASS	Index 35 is position/count 36; arithmetic remains 35.
V05 D/B-to-R cut-and-unroll morph	wait the geometrical way a D becomes PH.pdf; Tom Klootwijk...pdf pp. 1-4	PASS-AS-ADAPTER	Valid design transform; not a topology invariant or automatic physical process.
V06 Torque / phase / parity hinge	wait the geometrical way a D becomes PH.pdf pp. 2-8	PASS-BOUNDED	Parity toggles route/sheet in tests while continuous state remains explicit.
V07 Signed-distance primitives and CSG	KC Vector Graphical Art System(1).pdf pp. 1-3	PASS	Implemented and tested; analytic distance quality depends on primitive/operator.
V08 Log-polar coordinate LUT	KC Vector Graphical Art System(1).pdf; waarom-heeft-19...pdf pp. KC 1-2, 7-9; 19 doc 10-12	PASS-WITH-GUARD	Works for $r>0$; singular center is clamped and must not be treated as ordinary space.
V09 One-bit LUT / jitter output	KC Vector Graphical Art System(1).pdf pp. 1-10	PASS-AS-PROJECTION	1024x1024 packed payload is 131,072 bytes; CPU adapter is not the authoritative substrate.

ID / mechanism	Source pages	Status	Verified result
V10 Feynman-vector language	KC Vector Graphical Art System(1).pdf pp. 2-3, 7-10	REWRITE	Useful rendering metaphor; no quantum-performance claim retained.
V11 Local spherical/radial-angular support	Chronological Synthesis...pdf; Spherical Throughput...pdf pp. Chronological 5-7; Throughput 3-6	PASS	~35M candidate tests/s at 1M batch median across two pinned runs.
V12 Compatibility / double-vacuum separation	Chronological Synthesis...pdf; Spherical Throughput...pdf pp. Chronological 6; Throughput 3,7	PASS-BOUNDED	~76-86M compatibility tests/s at 1M batch; no literal second physical vacuum asserted.
V13 Event surface and guard crossing	Chronological Synthesis...pdf; Spherical Throughput...pdf pp. Chronological 5-8; Throughput 5-7	PASS-RESTRICTED	Sphere time error $\leq 3.91e-14$ s in synthetic cross-check; general relations remain conditional.
V14 Direct state_at(t) query	Chronological Synthesis...pdf pp. 7-8, 11-12	PASS-RESTRICTED	Direct cost exponent ~ 0.0009 ; 60 Hz replay exponent ~ 0.996 across two runs.
V15 Analytic next_event	Chronological Synthesis...pdf pp. 6-8, 11	PASS-RESTRICTED	~81M plane and ~14M sphere trajectory solves/s; only tested relation families are claimed.
V16 Transition routing / hourglass	Chronological Synthesis...pdf; hollowland-double-vacuum(9).pdf pp. Chronological 6-7; Hollowland 29-35	PASS-AS-ROUTING	Works as discrete routing; not evidence for literal spacetime pinch topology.

ID / mechanism	Source pages	Status	Verified result
V17 Möbius/Klein gluing	wait the geometrical way a D becomes PH.pdf; Chronological Synthesis...pdf pp. D/PH 5-10; Chronological 6	PASS-AS-ABSTRACT-TOPOLOGY	Roundtrip error 2.22e-16; no physical 3D self-intersection claim.
V18 Signed field on non-orientable topology	wait the geometrical way a D becomes PH.pdf; hollowland-double-vacuum(9).pdf pp. D/PH 7-10; Hollowland 12-35	CONDITIONAL-CORRECTION	Implemented with explicit orientation state; a globally single-valued sign is not assumed.
ID / mechanism	Source pages	Status	Verified result
V19 Lineage-based identity	Chronological Synthesis...pdf pp. 5-8, 11-12	PASS-WITH-PROBABILISTIC-LIMIT	~283M IDs/s; zero observed collisions is not a mathematical collision-proof guarantee.
V20 Irreducible external event log	Chronological Synthesis...pdf pp. 5-8, 11-12	PASS	Binary log ~1.3-1.5M events/s and 40 B/event; JSONL ~217 B/event in test schema.
V21 Spherical throughput metric	Spherical Throughput...pdf pp. 3,6,9-12	PARTIALLY-MEASURED	Compute-only median ~13.0k verified events/s for the specified 1M-candidate synthetic workload; not optical throughput.
V22 Hollowlens / waveguide liquid substrate	Spherical Throughput...pdf pp. 2-12	UNMEASURED-HARDWARE	No physical device was available; no loss, joules/event, settling time, or drift numbers are fabricated.
V23 Universal $O(1)$, zero memory/latency/power	Chronological Synthesis...pdf; Spherical Throughput...pdf pp. Chronological 1,8,11-12; Throughput 2,10	REJECT-UNIVERSAL	1M packed states consume ~104 MB; 1M state output adds another ~104 MB; costs scale with candidates/expression/event history.
V24 Physical self-forming Klein bottle / matter-as-code	Tom Klootwijk...pdf; george W bush.pdf; wait the geometrical way a D becomes PH.pdf pp. approx. 9-18 in overlapping dialogue sections	REJECT-AS-VERIFIED-CLAIM	Retained only as historical motif; no physical validation.
V25 Rendering as downstream projection	Chronological Synthesis...pdf pp. 1,5,9-10,12	PASS	Core tests run headlessly; projection has its own performance and quality limits.

4. Executable verification and coverage

34	2	93%	9
test cases	independent runs	coverage	contract gates
deterministic + randomized	different hash seeds	statement + branch	all pass

Test scope

- Batch state evaluation agrees with scalar formulas and does not mutate source state.
- Spherical support and compatibility batch predicates agree with independent scalar implementations over randomized inputs.
- Plane and sphere analytic event roots satisfy the relation at the computed hit point.
- A separate vectorized 48-iteration bisection path cross-checks analytic sphere entries.
- SDF CSG signs and operators are checked on known points.
- Möbius and Klein orientation-reversing loops satisfy two-loop roundtrip properties.
- Lineage generation is deterministic; one million generated IDs produced zero observed collisions.
- Binary event records round-trip through persistent storage.
- The complete pipeline preserves parity/sheet transition invariants and produces deterministic hashes.
- The one-bit projection is finite at the log-polar center and deterministic for a fixed seed.

Coverage summary

Area	Statements	Miss	Branches	Partial	Coverage
Package total	739	30	80	24	93%
Core topology	32	0	0	0	100%
Relations	103	3	12	3	95%
Compatibility	37	1	8	1	96%
Projection	36	1	2	1	95%
Pipeline	68	7	12	6	84%

Coverage interpretation

Coverage indicates exercised code paths, not proof of correctness. It is paired here with analytic residual checks, independent scalar comparisons, randomized topology properties, deterministic hashes, and two benchmark processes.

5. Practical performance metrics

Benchmark boundary

Field	Value
CPU	AMD EPYC 9V74 80-Core Processor
Affinity	CPU 0 only; one process
Visible logical CPUs	5
Memory	5.93 GiB
Python	3.13.5 (main, Jul 15 2026, 20:25:40) [GCC 14.2.0]
NumPy	2.3.5
Numeric threads	OMP/OpenBLAS/MKL/NumExpr = 1
Accelerators	No GPU, JIT, native extension, network, or energy meter
Aggregation	Median of two complete process-level benchmark runs

All performance numbers are local measurements from this container. They are useful for feasibility and scaling, not as universal machine-independent claims. Candidate arrays are synthetic and deterministic; event distributions and pruning ratios therefore describe the declared workload.

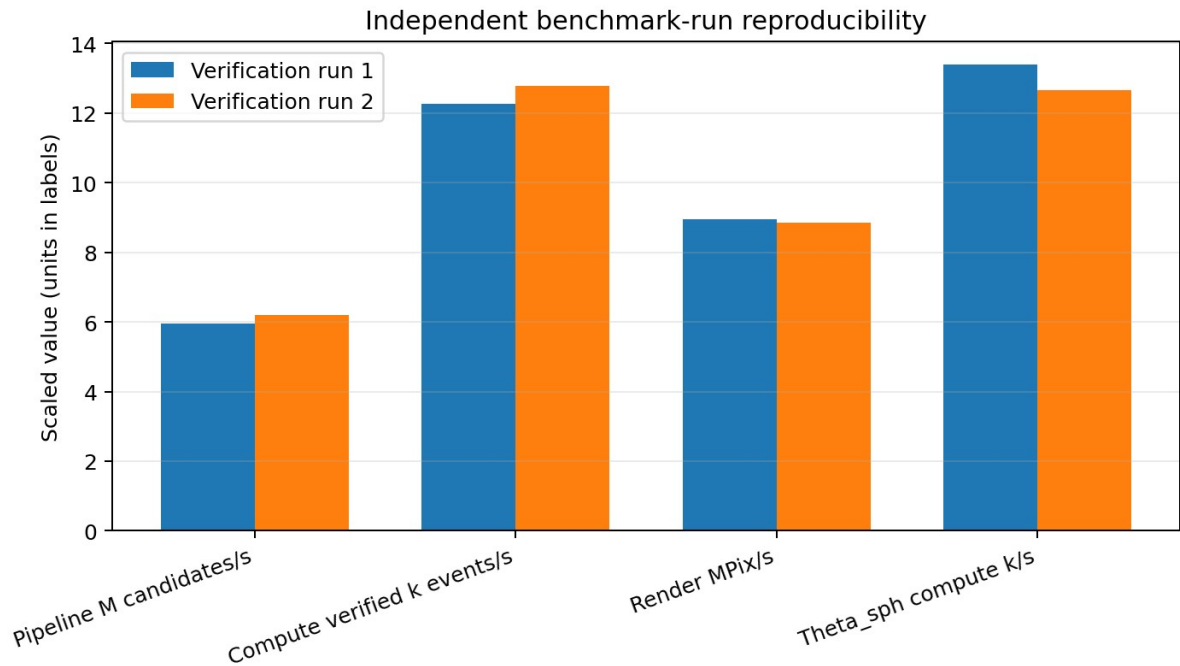


Figure 2. Two independent pinned single-core runs. Key throughput variation was approximately 1-9%, depending on operator and allocation pressure.

5.1 State query performance

A batch of one million packed states was evaluated at a horizon of 1,000,000.125 seconds in 122.19 ms median, or 8.20 million states/s. The two-run relative range was 8.55%.

States	Median ms	Aggregated p99 ms	Million states/s
1,000	0.084	0.134	11.94
10,000	0.906	1.109	11.04
100,000	9.725	9.894	10.28
1,000,000	122.188	143.856	8.20

Horizon independence result

The fitted scalar direct-query exponent was 0.0009; the 60 Hz replay exponent was 0.9963. This is the precise bounded form of the corpus's $O(1)$ -like claim: fixed expression cost can be independent of skipped frame count. It says nothing about unrestricted grammar growth, branch history, or exogenous events.

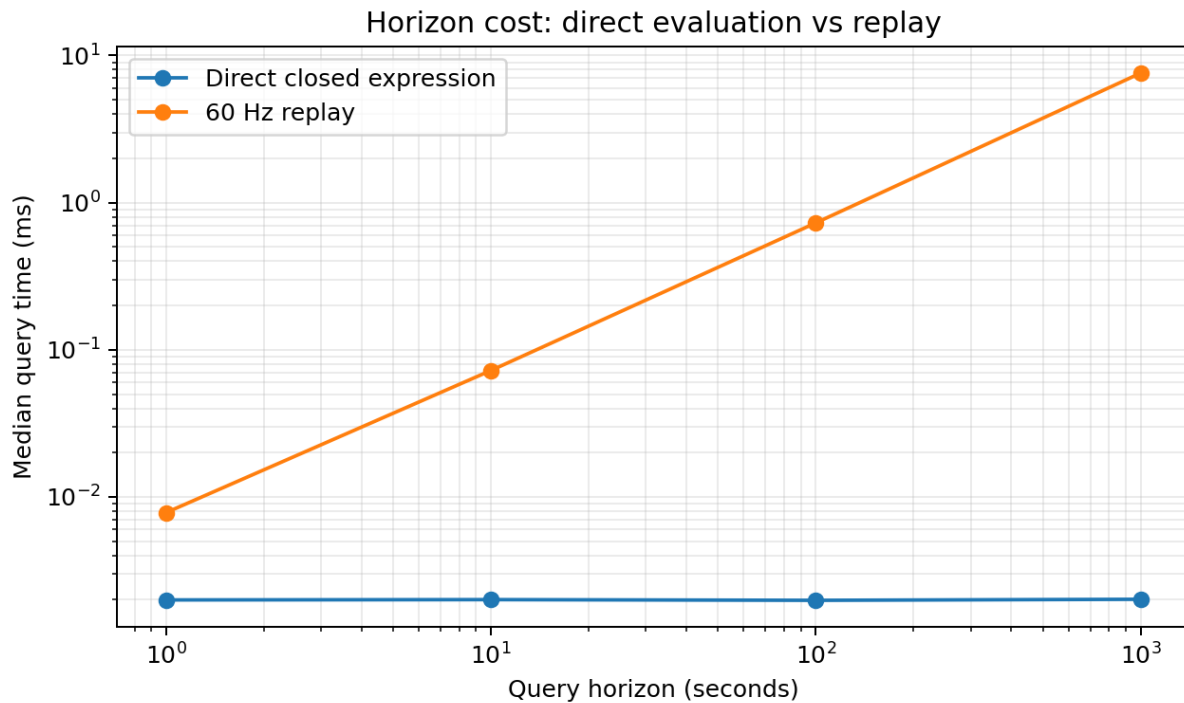


Figure 3. Direct closed expression versus sequential 60 Hz replay, 1-1000 second horizons.

5.2 Support, compatibility, and event solving

Operator, 1M items	Median ms	Million items/s	Scope
State evaluation	122.19	8.20	packed expression evaluation
Spherical support	28.26	35.40	7.00% admitted
Compatibility	13.18	76.38	1.20% accepted
Plane event solve	12.61	79.71	analytic linear root
Sphere event solve	73.27	13.65	stable quadratic root
Möbius route	7.54	133.25	orientation double cover
Klein route	12.33	81.07	quotient gluing
Lineage hash	3.87	259.65	64-bit deterministic mix

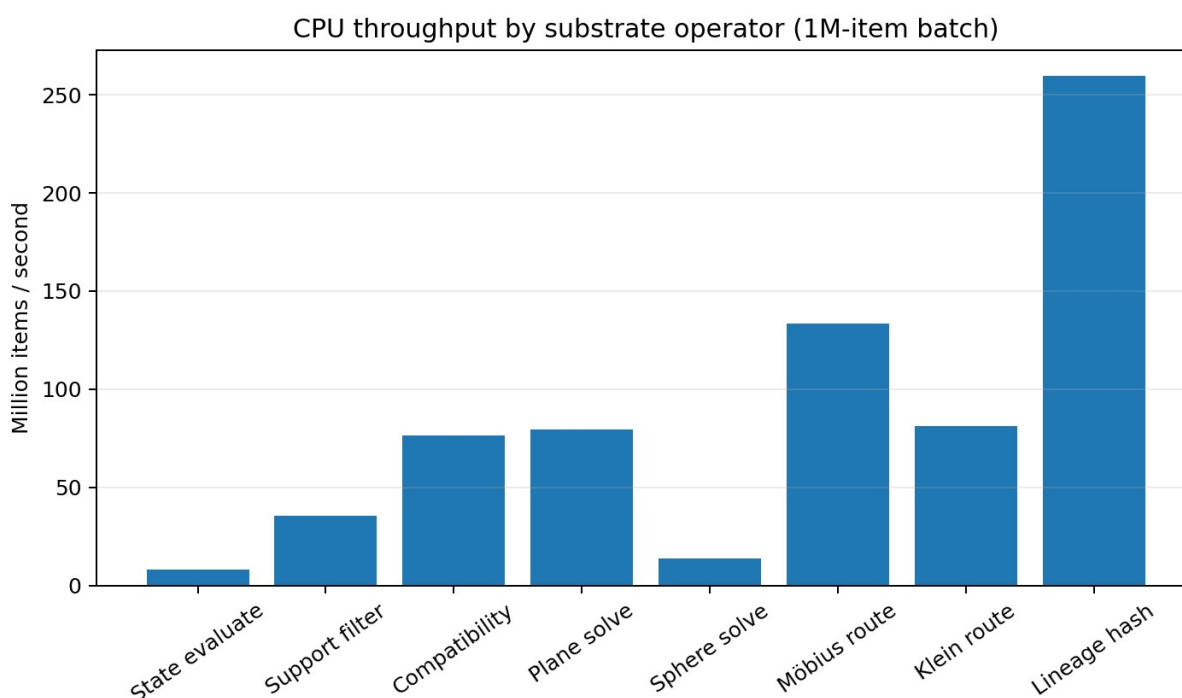


Figure 4. CPU operator throughput. These kernels have different arithmetic and allocation costs; bars are not interchangeable workload claims.

Independent event-solver cross-check

For 200,000 guaranteed sphere entries, the analytic quadratic solution was compared against a separate 48-iteration vectorized bisection path. Maximum absolute time error was $3.91\text{e-}14$ s; p99 error was $7.11\text{e-}15$ s; maximum relation residual was $1.22\text{e-}14$.

Meaning

The analytic solver is both faster and more accurate for this relation family. The bisection path verifies it independently and represents the slower fallback class required for generic bracketed relations.

5.3 Why event solving matters: brief-event stress test

A synthetic stress population of 250,000 sphere transits was generated with a median event duration of approximately 4.95 ms. Regular frame sampling declared an event only when at least one sample landed inside the sphere. Analytic roots had a known positive discriminant for every generated trajectory.

Sampler	Miss rate	Median delay ms	p99 delay ms	Missed / 250k
60 Hz	55.32%	5.58	16.36	138,301
120 Hz	40.96%	3.16	8.21	102,393
240 Hz	27.23%	1.71	4.11	68,067
1000 Hz	4.97%	0.48	0.99	12,436

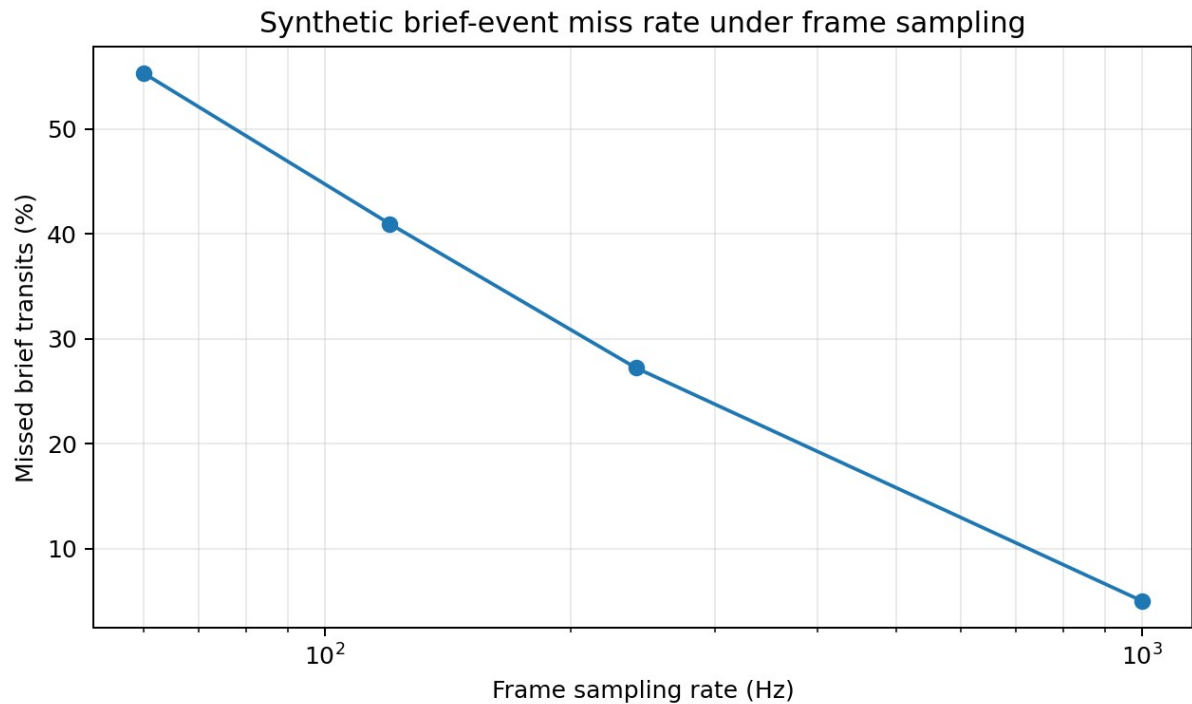


Figure 5. Stress-specific brief-event miss rates. This does not predict a general game’s event distribution; it demonstrates the failure mode of discrete frame sampling.

Accuracy boundary

The analytic result is not “zero tunneling” in a universal sense. It catches this declared non-degenerate relation family. Tangencies, multiple roots, discontinuities, moving relations, and unbracketed generic SDFs still require certified numerical policy.

5.4 End-to-end restricted pipeline

The measured pipeline performs state evaluation, spherical support, compatibility, plane root solving, horizon filtering, residual evaluation, and immutable transition-record creation. It scans all input candidates; no spatial index or GPU acceleration is present.

Input	Median ms	p99 ms	M candidates/s	Admitted	Compatible	Verified
1,000	0.218	0.259	4.60	136	13	5
10,000	1.274	1.354	7.85	1,229	59	26
100,000	12.312	12.803	8.12	11,787	453	187
500,000	75.917	76.562	6.59	59,963	2,351	998
1,000,000	164.859	227.647	6.07	119,815	4,820	2,063

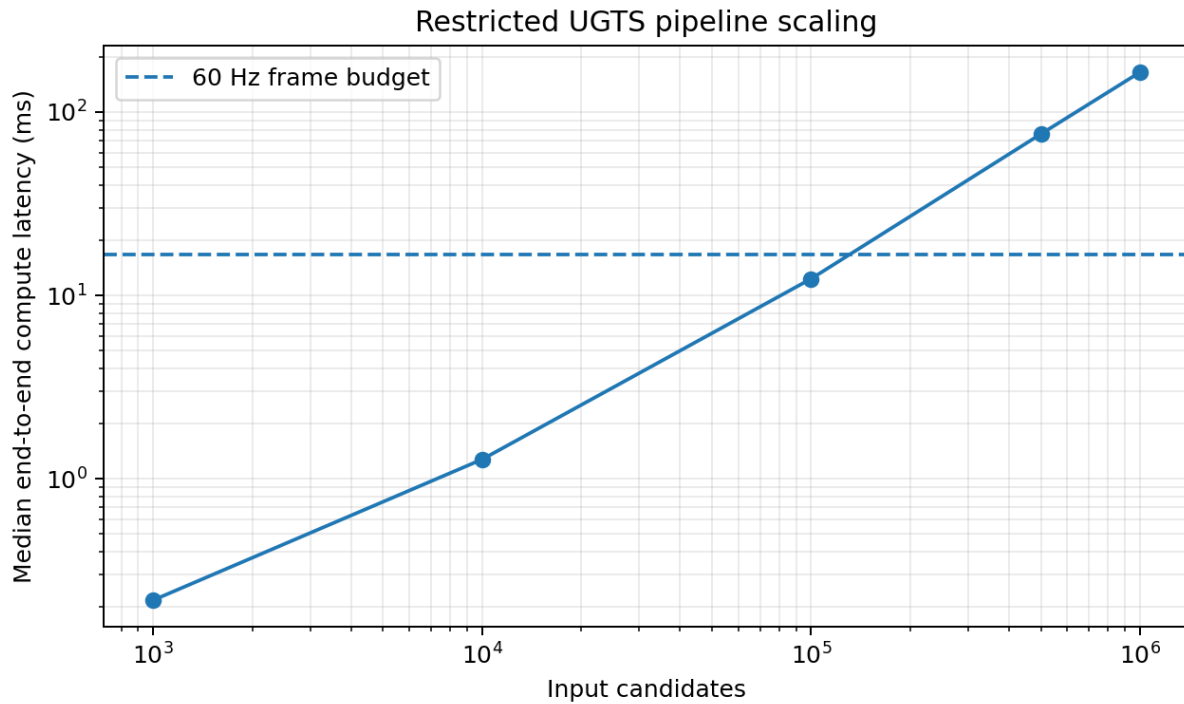


Figure 6. Approximately linear candidate scaling. The 100k workload fits the 16.67 ms reference line in this environment; 500k and 1M do not.

Practical interpretation

- 100k candidates: 12.31 ms median and 12.80 ms p99, with 187 verified events.
- 1M candidates: 164.86 ms median and 6.07M candidates/s, with 2063 verified events.
- The 1M support pruning ratio was 8.35:1; compatibility added 24.86:1 among admitted inputs.
- Five identical 200k runs produced one unique event hash, confirming deterministic output for the tested path.
- The next optimization priority is not exotic topology. It is broad-phase indexing, compiled kernels, reuse of temporary arrays, and parallel scheduling.

6. Topology, lineage, event logs, and projection

6.1 Topology as quotient routing

One million Möbius coordinates were wrapped at 133.25 million coordinates/s; Klein quotient routing achieved 81.07 million/s. A randomized two-loop roundtrip over 250,000 points had maximum coordinate-or-bit error 2.22e-16.

This verifies the information-architecture role of non-orientable topology: routing can reverse orientation while remaining deterministic. It does not verify a physical non-self-intersecting 3D embedding, self-assembly, or global signed distance without an orientation cover.

6.2 Lineage and persistence

The vectorized lineage generator produced one million 64-bit IDs at 259.65 million IDs/s with zero observed collisions. The result is probabilistic: a 64-bit hash cannot provide mathematical collision impossibility. Production identity should include collision detection, larger digests, or a registry policy.

Event-log operation	Throughput	Density
Binary write	1.58M events/s	40 B/event
Binary read	1.27M events/s	40 B/event
JSONL write	218k events/s	217.1 B/event

The test JSONL schema consumed 5.43 times the binary payload. Binary is compact and fast; JSONL is inspectable and easier to evolve. A production system may use a framed binary format with schema/version/provenance metadata.

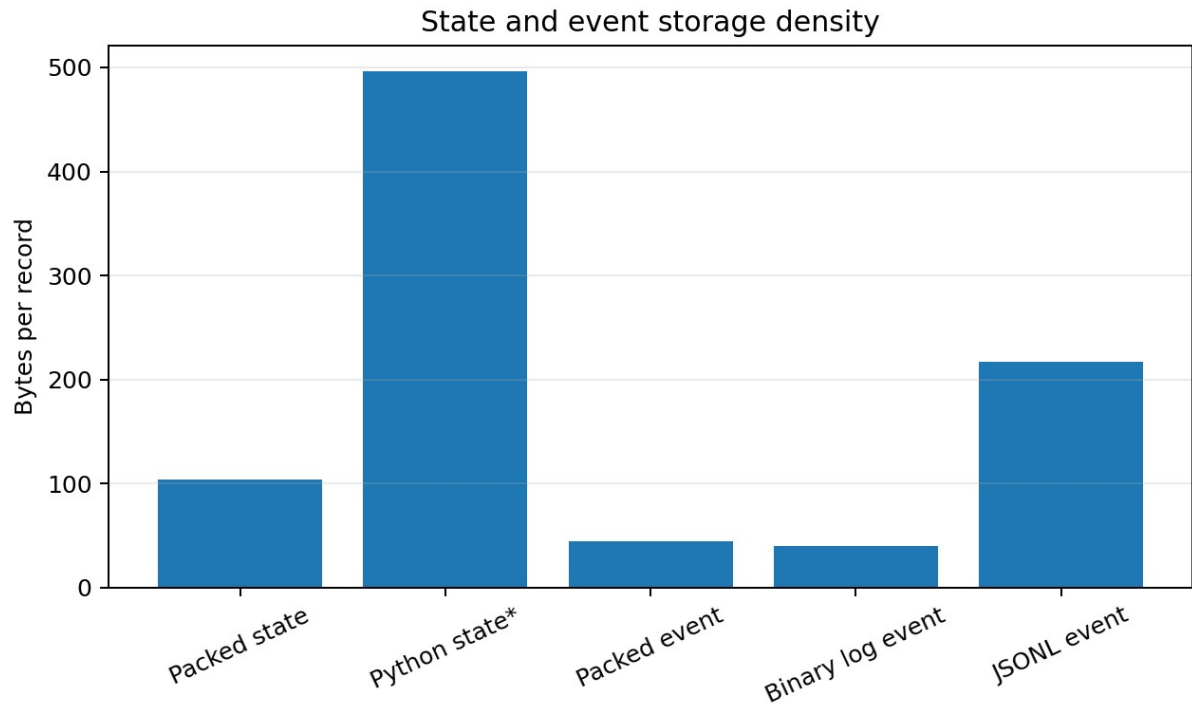


Figure 7. Record density. The Python object estimate includes recursively measured fields but still depends on allocator/runtime behavior.

6.3 Downstream log-polar SDF one-bit projection

The projection adapter is intentionally separate from state authority. It maps pixels to log-polar coordinates with a clamped center, applies a bounded angular warp, evaluates an equilateral-triangle SDF, converts edge coverage through deterministic one-bit jitter, and packs the result.

Resolution	Median ms	CPU fps	MPix/s	Packed bytes
256x256	5.96	167.8	11.00	8,192.0
512x512	24.05	41.6	10.90	32,768.0
1024x1024	117.76	8.5	8.90	131,072.0

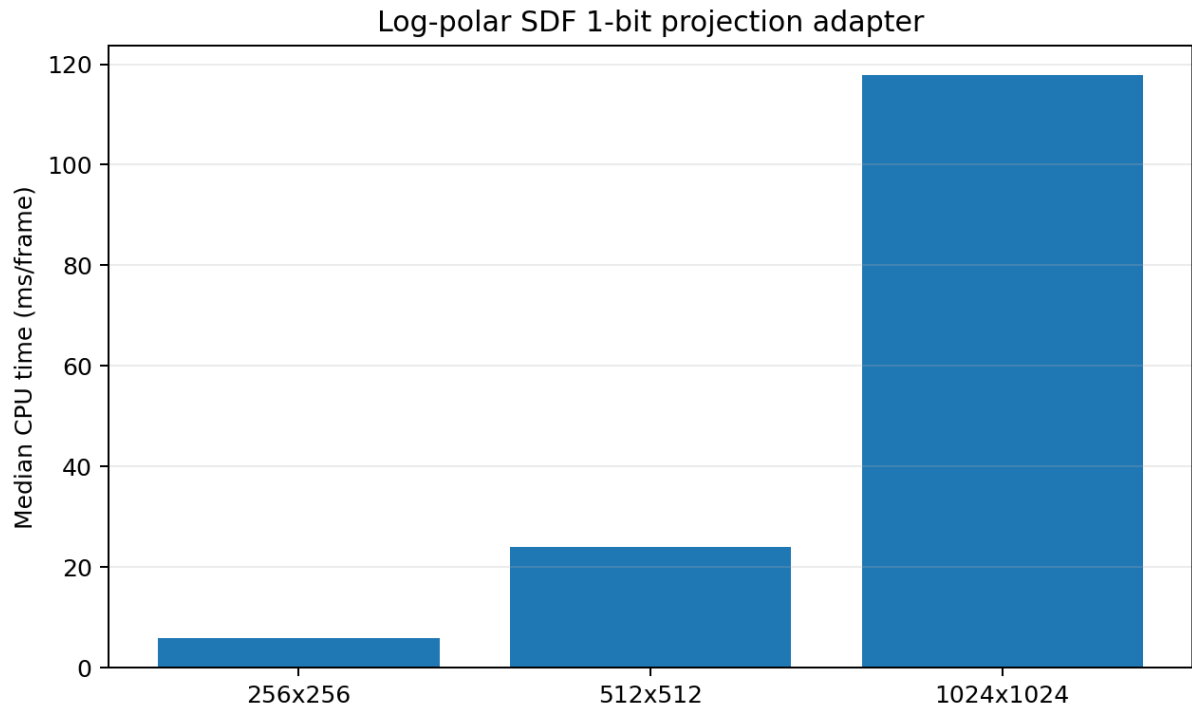


Figure 8. CPU-only projection cost. A GPU shader is the appropriate production path for high-resolution frame rates.

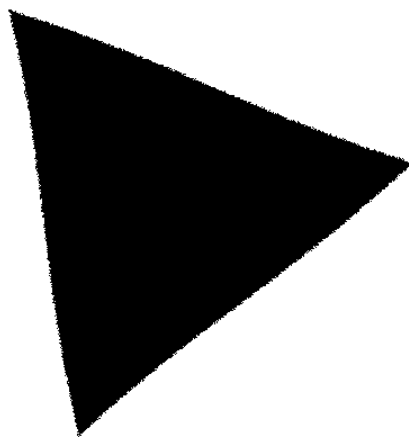


Figure 9. Deterministic 512x512 one-bit sample. Projection test only.

Rendering conclusion

The adapter is technically valid and compact, but the current NumPy CPU path reaches only about 8.5 fps at 1024x1024. It should be moved to GPU/compute or compiled SIMD if used in a real-time graphics product.

6.4 Practical metric register

Area	Metric	Value	Unit	Scope
Verification	Unit tests	34 tests, two hash seeds	all pass	test_run_hash0.txt; test_run_hash123.txt
Verification	Statement/branch coverage	93% total	coverage report	coverage.txt
State	Scalar direct state_at horizon exponent	0.0009	dimensionless slope	1-1000 s horizon
State	60 Hz replay horizon exponent	0.9963	dimensionless slope	1-1000 s horizon
State	1M packed state evaluation	8.20	million states/s	122.19 ms median
Filter	1M spherical support	35.40	million candidates/s	admitted 7.00%
Filter	1M compatibility	76.38	million candidates/s	accepted 1.20%
Event	1M plane roots	79.71	million trajectories/s	analytic linear crossing
Event	1M sphere roots	13.65	million trajectories/s	max residual 2.69e-12
Event	Analytic vs bisection time error	3.91e-14	seconds max	200k guaranteed entries
Pipeline	100k candidates	12.31	ms median	8.12M candidates/s
Pipeline	1M candidates	164.86	ms median	6.07M candidates/s; 2063 verified
Pipeline	Determinism	1	unique output hash	five identical runs
Accuracy	60 Hz brief-transit miss stress	55.32	percent missed	synthetic event durations median ~4.95 ms
Accuracy	1000 Hz brief-transit miss stress	4.97	percent missed	same synthetic distribution
Topology	Möbius roundtrip / routing	133.25	million coordinates/s	roundtrip error <= 2.22e-16
Topology	Klein quotient routing	81.07	million coordinates/s	abstract gluing map
Lineage	1M lineage IDs	259.65	million IDs/s	0 observed collisions
Log	Binary event write	1.58	million events/s	40 bytes/event
Log	Binary event read	1.27	million events/s	40 bytes/event

Area	Metric	Value	Unit	Scope
Log	JSONL event write	218	thousand events/s	217.1 bytes/event
Memory	Packed state	104	bytes/entity	1M array RSS delta ~104 MB
Memory	Packed event	44	bytes/event	binary persistent record is 40 bytes
Render	256x256 log-polar SDF 1-bit	5.96	ms/frame	167.8 fps CPU
Render	512x512 log-polar SDF 1-bit	24.05	ms/frame	41.6 fps CPU
Render	1024x1024 log-polar SDF 1-bit	117.76	ms/frame	8.5 fps CPU; 131072 packed bytes
Throughput	Theta_sph compute-only	13026	verified events/s	software workload only; R_support 8.34, R_compat 25.04

6.5 Spherical-throughput accounting

The source defines throughput as verified outputs, not raw detector or candidate traffic. The software benchmark therefore counts only records that pass support, compatibility, an event root, horizon, and residual checks.

```
Theta_sph = N_verified / elapsed_time

N_verified = sum( support AND compatibility AND guard_crossed AND confidence_ok )
```

13.03k	8.34:1	25.04:1	1.69%
compute verified events/s software workload	R_support input / admitted	R_compat admitted / compatible	eta_route verified / admitted

This compute-side figure excludes the physical source, optical coupling, liquid actuator, detector, ADC, FPGA/MCU, calibration, and energy boundary. It must not be presented as Hollowlens throughput. The hardware source explicitly requires those terms to be measured beside the verified event rate.

Hardware boundary

The only honest hardware metric currently available is “not measured.” The package therefore includes a preregistration template rather than invented values.

7. Hardware measurement boundary

Hollowlens-0 status

The uploaded waveguide document supplies a practical architecture: bounded spherical input, tunable liquid coupling, compatible guided mode, measured guard crossing, verified output. It also states that component precedent is not system validation. No physical prototype was supplied in this task.

Required bench metrics

Metric	Definition	Unit
Theta_sph	verified events / wall second	events/s
eta_route	verified / admitted	ratio
Theta_E	verified events / total system energy	events/J
R_support	inputs / supported	ratio
R_compat	supported / solver-entering	ratio
L50 / L99	median / tail input-to-event latency	s
P_miss / P_false	missed / false event probability	probability
Insertion loss	input-to-useful detector loss	dB
Settling time	actuator command to stable transfer	s
Drift	transfer error vs time and temperature	specified error/time/K
Calibration burden	time and energy per operating interval	s and J
Cycle life	cycles before failure/out-of-tolerance	cycles

Preregister before power-up

- Wavelength, optical power, angular/radial support, accepted modes, thresholds, confidence rule, and uncertainty boundary.
- Complete energy boundary: source, actuator, detector, ADC, FPGA/MCU, control computer, and calibration amortization.
- Electronic reference implementation at equal error and uncertainty.
- Temperature, pressure, fluid composition, packaging state, bubble/contamination checks, and calibration interval.
- Kill criteria: false events, loss, slow actuation, drift, calibration burden, packaging failure, or no end-to-end advantage.

Go/no-go test

Build Hollowlens-0 only if the complete system produces more verified events per second and per joule than the matched electronic reference at the same error boundary. Component existence is not enough.

8. Missing work, risks, and kill criteria

What is still missing

Gap	Required next step
Grammar closure	No general compiler proves that composed rules remain inside a bounded, normalized relation family.
Broad phase	The reference scans candidates linearly. Spatial/semantic indexes are required for larger worlds.
Numerical certification	Tangencies, coincident surfaces, singular centers, rank changes, and long horizons need intervals or set-valued answers.
Transitions	Production transitions need relation-specific invariants, rollback, conflict handling, split/merge lineage, and schema evolution.
Parallel execution	No GPU, native SIMD, task scheduler, distributed event ordering, or network replication benchmark exists.
Game integration	No Unity/Godot/Unreal frame, rollback, physics, or ECS comparison was run in this verification.
Projection quality	No perceptual error, temporal flicker, print dot-gain, or anti-aliasing quality metric was measured.
Physical hardware	No optical loss, energy, settling, drift, bubbles, calibration, or reliability data exists.
Semantic/agent layer	A shared relation language is not by itself cognition, general AI, biological explanation, or consciousness.

Software kill criteria

- Compatibility plus event solving grows as fast as or faster than the explicit structure it was intended to avoid.
- Grammar composition causes uncontrolled expression, root, or branch explosion.
- Certified event ordering cannot be maintained at application tolerance.
- Lineage cannot reconstruct split/merge histories without ad hoc global bookkeeping.
- The reference becomes slower, larger, or less accurate than a conventional indexed event engine for the same query contract.
- Projection work leaks back into the authoritative substrate and recreates hidden frame dependence.

Hardware kill criteria

- Required event rate exceeds liquid/actuator bandwidth.
- Optical loss or crosstalk prevents the declared error target.
- Calibration energy/time erases any throughput or energy advantage.
- Bubbles, evaporation, leakage, hysteresis, swelling, or alignment drift prevent repeatability.
- The digital sidecar performs nearly all useful work and the optical kernel adds no measured system advantage.

9. Recommended next build

UGTS-2 target

Do not add more metaphors. Add a compiled broad phase, certified event brackets, explicit transition schemas, and a matched baseline benchmark.

Software milestone

3. Implement an indexed 2D Equation World Zero: 1M stored states, but only locally supported relation families enter the event solver.
4. Add uniform-grid/BVH or semantic support indexes and compare against the present linear scan.
5. Compile plane/sphere and selected polynomial relations into vectorized kernels; reuse temporary arrays.
6. Add interval guards for tangencies and uncertain coefficients; return intervals instead of false precision.
7. Implement lineage split/merge policies and a versioned binary event-log schema.
8. Benchmark `state_at`, `next_event`, `events_in_support`, routing, replay/rebuild, and rollback against a conventional engine baseline.
9. Only after the headless contract is stable, add GPU projection and a minimal game-engine adapter.

Physical milestone

The physical program should remain separate: one scanned input, one tunable liquid coupler, one 2x2 interferometer, balanced detection, and a digital sidecar. Preregister every metric in the included hardware template. Stop if full-system energy, latency, accuracy, and calibration do not beat the reference.

Final decision

The substrate has passed a meaningful bounded verification. It is no longer only a collection of geometric associations: a technically coherent kernel exists, its restricted claims are executable, and its practical limits are measurable. The next value comes from indexing, certification, compiled execution, and matched baselines—not from widening the ontology before the current one is stressed.

Appendix A. Source and build manifests

A.1 Uploaded source PDFs

Ref	Filename	Pages	SHA-256 prefix
S1	21BenBurgersStrikesBackTelNetNet(2).pdf	19	1ab4f04669ded1aa...
S2	Chronological Synthesis of the Spherical Substrate Line(5).pdf	12	9bacfe071a64951f...
S3	KC Vector Graphical Art System(1).pdf	10	6518f7f73f46ec6c...
S4	Spherical_Throughput_Practical_Waveguide_Liquid_Substrate_Lensing(sqwit (super quick wit)(2).pdf	12	949d7ee3cd7c9e4f...
S5	Tom Klootwijk NL200678942 10-07-1990.pdf	52	543515e2a0f3b545...
S6	george W bush.pdf	42	41fab10296500bfc...
S7	hollowland-double-vacuum(9).pdf	36	69a6b341d02c87f4...
S8	waarom-heeft-19-drie-lettergrepen-en-in-binaire-waarom-het-kantelpunt-tussen-1-en-2-cijfers-geometrisch.pdf	30	745ceb1251ac496d...
S9	wait the geometrical way a D becomes PH.pdf	34	66f17de45fd6e358...

Exact full hashes are in `results/source_manifest.json`. The source PDFs are not duplicated inside the final ZIP; the package contains only the derived verification, implementation, and manifests.

A.2 Build contents

Path	Purpose
<code>src/ugts</code>	Reference state, support, compatibility, relation, solver, topology, lineage, log, render, and pipeline modules.
<code>tests/test_ugts.py</code>	34 unit and randomized tests.
<code>benchmarks</code>	Two-run benchmark, aggregation, chart generation, and contract verification.
<code>results</code>	Raw run JSON, aggregate CSV/JSON, test logs, coverage, verification matrix, manifests.
<code>docs</code>	Technical specification, limitations, source traceability, metric summaries, hardware template.
<code>examples</code>	Headless pipeline, topology, and one-bit projection examples.
<code>report</code>	This report in DOCX and PDF.

A.3 Reproducibility checks

Contract gate	Result
two completed runs	PASS
direct horizon exponent below 0 1	PASS
replay horizon exponent near linear	PASS
sphere crosscheck below 1e 10 s	PASS
topology roundtrip below 1e 12	PASS
deterministic output	PASS
pipeline scaling approximately linear	PASS
state layout 104 bytes	PASS
binary log 40 bytes	PASS

A.4 Verification environment

Field	Value
CPU	AMD EPYC 9V74 80-Core Processor
Visible logical CPUs	5
Pinned affinity	0
Platform	Linux-6.18.35-x86_64-with-glibc2.41
Python	3.13.5 (main, Jul 15 2026, 20:25:40) [GCC 14.2.0]
NumPy	2.3.5
Thread limits	OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1, MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1
Scope	single Python process; CPU-only; no GPU; containerized KVM host

Appendix B. Reproduction and definitions

B.1 Reproduction commands

```
python -m venv .venv
. .venv/bin/activate
pip install -r requirements.txt

PYTHONPATH=src python -m unittest discover -s tests -v
coverage run --branch -m unittest discover -s tests -v
coverage report -m

./benchmarks/run_all.sh
python benchmarks/aggregate_runs.py
python benchmarks/make_charts.py
python benchmarks/verify_contract.py
```

B.2 Metric definitions

Metric	Definition
Candidate throughput	Input records processed divided by measured compute time.
Verified event	A record passing support, compatibility, root/horizon, and residual checks.
Theta_sph compute	Verified software events divided by measured pipeline compute time. Not optical throughput.
R_support	Input candidates divided by admitted support candidates.
R_compat	Admitted candidates divided by compatibility-passing candidates.
eta_route	Verified events divided by admitted inputs in the declared workload.
Median / p99 latency	Median or 99th-percentile measured function wall time across benchmark repetitions, then aggregated across two process runs.
Scaling exponent	Slope of $\log(\text{time})$ versus $\log(\text{problem size or horizon})$.
Residual	Absolute relation value at the computed event point.
Roundtrip error	Maximum coordinate or orientation-bit difference after the declared quotient loops.

B.3 Source references used as architectural anchors

Source	Use in this verification
[S2] Chronological Synthesis of the Spherical Substrate Line	pp. 1, 5-12: query/event interpretation, bounded feasibility, complexity, risks, and prototype program.
[S4] Spherical Throughput Practical Waveguide Liquid-Substrate Lensing	pp. 2-12: implementable operator rule, verified-event metric vector, hardware boundary, kill criteria, and component-vs-system distinction.
[S3] KC Vector Graphical Art System	pp. 1-10: log-polar, SDF/CSG, one-bit output, display/prepress adapter motifs.
[S8] Why 19...	pp. 1-10: radix thresholds, popcount, Pascal parity, and Sierpinski procedural motif.
[S1/S5-S7/S9] Dialogue sources	Geometric morphs, parity hinge, double-vacuum, hourglass, Möbius/Klein, COBOL/schema, and speculative physical extensions; retained only where technically bounded.

Closing statement

This report verifies a restricted executable substrate and identifies the exact boundary where evidence stops. The package is designed so the next experiment can fail informatively rather than succeed rhetorically.